Wai Ka (Bella) Sun | Curriculum Vitae

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Education

- **Tsinghua University** — Beijing, China
  Bachelor of Engineering in Architecture — Expected July 2026
  - **GPA:** 3.8/4.0 **Rank: 1/7** (School of Architecture, Class of 2026)
  - **Relevant Graduate Coursework:** Brain-Computer Interface & EEG Signal Processing (Grade: 4.0), Minimally Invasive Brain Machine Interfaces (Grade: 4.0), Introduction to AI Industry (Grade: 4.0), Interaction AI (Grade: 4.0), Neural Modeling and Data Analysis (In progress), Brain and Machine Intelligence (In progress).
- **Delft University of Technology (TU Delft)** — Delft, Netherlands
  Exchange Student, AI & Digital Twin Innovation Program — Aug 2022 – Feb 2023
  - Research Focus: Automated algorithms for digital twin urban planning.

Research Interests

Key Interests: Human-centered AI, Brain-Computer Interfaces (BCI), Robotic Learning, and Computer Vision applied to Assistive Technology and Ambient Intelligence; intersecting with Computational Neuroscience to decode human intent.

Honors & Awards

- **Mingxi Scholarship** (2026) — ***4/533 Awarded*** *by the Chief Executive of Hong Kong*
- **Special Scholarship, Ministry of Education of China** (2021) — ***Highest*** *undergraduate honor in HK, China*
- **Rhodes Scholarship Finalist** (2025) — *Selected for final interview(s)*
- **Tsinghua Disruptive Innovation Talent Program Grant** (2024) — *First batch selected for 6-figure research funding*
- **Comprehensive Excellence Award** (2021, 2024, 2025) — *Awarded to* ***Top 5%*** *of students at Tsinghua University*
- **National Undergraduate Innovation Annual Conference Reform Award** (2023) — *Sole undergraduate project*
- **Annual Excellent Design Prize** (2024) — *Awarded by Tsinghua School of Architecture (Project: Pipedream)*
- **Ng Teng Fong Hong Kong Student Inspirational Scholarship** (2023) — *First recipient for sci-tech innovation*
- **Luoyuehua Scholarship** (2024, 2025) — *For outstanding female students promoting STEM inclusivity*
- **Champion**, Beijing University Sailing Regatta (2025)

Research & Project Experience

Multimodal Semantic Analysis for Architectural Schematics — *Thesis Researcher*
Comprehensive Undergraduate Thesis, Tsinghua University — Jan 2026 – Present
- **Achievement:** Pioneering a multimodal AI retrieval and analysis framework to bridge the "semantic gap" in architectural drawings, aligning AI capabilities with professional design logic (spatial scales, tectonic logic, drawing types).
- Developing a system to transition architectural workflows from traditional CAD/BIM modeling to AIGC-assisted conceptual design, optimizing spatial object retrieval for complex environments.

Visual Perception Mechanisms in Polar Mimicry Environments — *Independent Researcher*
Advisor: Prof. Wang Hui (THU) — Sept 2025 – Dec 2026
- **Achievement:** Quantified "whiteout" sensory deprivation and cognitive decay, correlating environmental monotony with physiological glare reaction latency using high-fidelity 14-channel EEG data.
- Constructed a high-fidelity naked-eye 3D polar simulation in Unreal Engine 5.4 within an n'Space VR cabin to study human-factor engineering and neuro-visual stimuli in extreme conditions.

Interpretable & Robust Humanoid Control: Whole-Body Control for Unitree G1 — *Independent Researcher*
Advisor: Prof. Mengdi Xu (IIIS, Tsinghua University) — Sept 2025 – Jan 2026
- **Achievement:** Verified CfC-based Neural Circuit Policies (NCPs) via PPO on humanoid benchmarks, achieving a ~52% parameter reduction compared to MLPs while significantly enhancing interpretability and robustness against sensor noise.
- Built a Sim-to-Real motion tracking pipeline mapping AMASS data to a 29-DoF **Unitree G1** humanoid in Isaac Sim.
- Modified URDF to add **3-DoF torso articulation** (Pitch, Roll, Yaw), enabling high-fidelity tracking of trunk rotations and rhythmic nonlinear movements.

MindBridge: Generative AI & BCI Interaction System *Project Lead*
Advisors: Prof. Gao Xiaorong (THU), Prof. Liu Yunhao (ACM & IEEE Fellow, Academician of CAS) Apr 2024 – Present

- **Achievement:** Evaluated anonymously by academicians and selected as one of the **first** batches of disruptive innovation talent training programs at Tsinghua University, securing **6-figure pre-seed funding**.
- Initiated a non-invasive BCI system to decode user intent for environmental control (e.g., smart home, robotics).
- Engineered a Python pipeline using FBCSP to extract features from micro-EEG signals (Motor Imagery).
- Integrated Large Language Models **(LLMs)** to translate abstract neural classifications into complex commands.

Babel Tower Challenge: Sim-to-Real Robotic Control *Lead*
Advisor: Prof. Zhao Qianchuan (Director of CFINS, Dept. of Automation) Spring – Summer 2025

- **Achievement:** Optimized control strategies for dynamic friction and gravity, achieving **3x baseline performance** in stacking tasks and successfully bridging the Sim-to-Real gap in a **5-DOF robotic arm** competition.
- Implemented DAgger (Dataset Aggregation) to aggregate expert demonstrations, correcting covariate shifts in RL policies.

AIUI: Future HCI Architecture via Long-Term Dialogue *Research Assistant*
Advisor: Prof. Gong Jiangtao (THU), Tsinghua Institute for AI Industry Research (AIR) Dec 2024 – Mar 2025

- **Achievement:** Implemented a **Retrieval Augmented Generation (RAG)** architecture to incorporate spatio-temporal context, significantly reducing cognitive errors in multi-round agent dialogues.
- Developed a multimodal agent framework using LLMs to supersede traditional GUIs with intent-driven conversation.

TwinSpace: XR Environment Remodeling *Project Lead*
Advisor: Prof. Liu Yunhao (ACM & IEEE Fellow, Academician of CAS) Nov 2021 – Dec 2023

- **Achievement:** Selected as the sole undergraduate project for the **China National Innovation Annual Conference**.
- Designed a zero-threshold XR interaction system using gesture recognition and self-trained AIGC models.
- Constructed a CAVE-like immersive simulation and curated a massive pose dataset.

Pipedream: BCI-based Generative Design *Co-First Creator*
Advisors: Prof. Zhang Xin (THU), Jiajian Min (MYStudio), Anna Yu (MYStudio) Feb 2023 – Apr 2024

- **Achievement:** Featured in a **CCTV documentary** "Focus on Innovation"; awarded Annual Excellent Design Prize.
- Developed a digital twin somatosensory game integrating BCI (brainwave-interactive level branching) with AIGC.

Industry Experience

NetEase Fuxi AI Lab Hangzhou, China
AI Product Development Intern July 2023 – Nov 2023

- **Achievement:** Constructed **8 large-scale corpora** and test sets to guide iterative model R&D, significantly optimizing Agent decision-making logic and user behavior perception.
- Supported the productization and multi-modal application of the core Large Language Model **"Yuyan"** for **AI Agents** in the flagship MMORPG *Justice Online*.
- Strategized and tracked **30+ AI features**, focusing on robustness optimization and enhancing user interaction depth.

Publications & Selected Works

- **Sun, W.K.**, et al. "White Paper on Chinese Ice Sailing." (2025). Presented to the International Sailing Federation; established China's first international venue certifications.
- **Sun, W.K.**, Zhao, R. "Pipedream: BCI-based Generative Design." (2024). Featured Project in **CCTV documentary**.
- **Sun, W.K.**, et al. "TwinSpace: XR Environment Remodeling Based on Posture Recognition." (2023). Conference Presentation at the **China National Undergraduate Innovation Annual Conference**.

Leadership & Service

- **Founder & President**, Tsinghua University Sailing Association & Asia-Pacific Intercollegiate Sailing Federation.
- **Author**, "White Paper on Chinese Ice Sailing" (2025). Established standards for the International Sailing Federation.
- **Head of Research and Publication**, Youth League Committee (THU). Organized technical lectures and taught disadvantaged groups how to leverage **Generative AI** tools to develop side businesses and bridge the digital divide.
- **Young Envoy of UNICEF** (Secondary School). Advocated for children's rights.
- **Technical Director**, Student Science & Technology Association. Organized GenAI workshops for 100+ students.

Interests & Global Exploration

- **Sports:** Competitive Sailing (Inshore Cruising), Equestrianism, Paragliding, Ice-skating, Shooting, Snooker. Advanced Open Water (AOW) certified diver; 3m springboard diving 103C & platform diving 101B; Small aircraft pilot.
- **Expeditions:** **Super Backpacker** who has explored **38 countries** (27 independently). Survived a **blizzard** and **food shortage** during a 5-day solo expedition in **Greenland**, honing extreme resilience.
- **Slash Youth:** Poet & dancer; Organized Tsinghua's Modern Dance Showcase (2025). Published my first 3D FPS game in 2023. As a social media KOL & talk show host, I have **10k followers** across the entire network and **3M** views.

Technical Skills

- **Programming:** Python (PyTorch, NumPy, Scikit-learn), C#, MATLAB.
- **AI & Control:** Reinforcement Learning (PPO, DAgger), Generative AI, LLMs (RAG), ROS, Isaac Sim, MuJoCo.
- **Signal Processing:** EEG Analysis, FBCSP, SSVEP, Motor Imagery pipelines.
- **Design:** Unity 3D, Unreal Engine, Houdini, Blender, Rhino & Grasshopper.
- **Languages:** English (Fluent), Cantonese (Native), Mandarin (Native), Japanese (Beginning).